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Explainable artificial intelligence in LSTM transformer models for student performance analysis

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Abstract

Accurate prediction of student performance is essential for facilitating early interventions and enhancing academic results. However, interpretability is lacking in traditional machine learning approaches, which frequently struggle to capture implicit behavioural and learning dependencies within educational data. The study utilizes a publicly available dataset from the Kaggle repository, containing student records with features related to academic, demographic, and behavioural attributes. The dataset was standardized using z-score normalization. Four deep learning models were developed based on their effectiveness in processing sequential data: LSTM, CNN, RNN, and a hybrid model combining LSTM and Transformer architectures (LSTM-Transformer). To ensure model transparency, we integrated Explainable Artificial Intelligence (XAI) techniques such as SHapley Additive exPlanations (SHAP) and Transformer attention mechanisms. The hybrid LSTM-Transformer model consistently achieved the best performance, with the lowest MSE and MAE and highest R^2 across most runs. Statistical tests confirmed significant differences among models (Friedman $p < 0.0001$), with the hybrid outperforming CNN ($p = 0.0286$) and RNN ($p < 0.0001$). While LSTM and the hybrid showed comparable accuracy ($p > 0.05$). The hybrid model's interpretability and performance make it a strong candidate for future real-world deployment; however, further work is needed to assess integration into educational systems, computational feasibility, and stakeholder acceptance. According to SHAP and attention studies, absenteeism, study time, and parental support were identified as the most influential predictors of performance, while gender and parents' level of education had little bearing. This research presents a balanced framework that advances both accuracy and interpretability in student performance modelling.

Keywords Explainability, Hybrid model, Deep learning, Attention mechanism, Student performance prediction

1 Introduction

Predicting student performance is a critical area of educational research, as it enables timely interventions, informed policy-making, and the design of personalized learning strategies. However, in addition to accuracy, educational prediction systems must



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be interpretable to foster trust and responsible decision-making. Explainable Artificial Intelligence (XAI) has therefore become essential, enabling educators to understand, validate, and act on model predictions. Traditional machine learning (ML) approaches have been widely employed in this domain but often fall short in modeling the complex behavioral dependencies inherent in educational data and lack interpretability in their outputs [12, 25].

Deep learning methods such as Long Short-Term Memory networks (LSTM), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) have shown promise in handling high-dimensional and sequential data. LSTM models, for instance, are effective for capturing temporal trends in student performance [14], while CNNs excel in identifying patterns within structured data [26]. Despite their capabilities, these models often operate as black boxes, limiting their utility in educational decision-making where interpretability is essential [12].

Recent advances in hybrid architectures that combine LSTM with Transformer encoders offer an opportunity to enhance both predictive power and model transparency. These models benefit from LSTM's ability to capture sequential patterns and the attention mechanisms of Transformers, which highlight important features contributing to predictions [2]. Integrating explainability tools, such as SHapley Additive exPlanations (SHAP), further supports the interpretability required for real-world application in education [12]. This study aims to develop an interpretable deep learning framework for predicting student academic performance using a hybrid LSTM-Transformer architecture. The research also investigates the comparative performance of LSTM, CNN, and RNN models and applies explainability techniques to identify key academic and behavioural predictors influencing student outcomes. This study is guided by the following research questions:

1. How can LSTM – Transformer hybrid model be developed to predict student performance?
2. How can explainability techniques be integrated into the hybrid model to interpret and explain its predictions?
3. What are the key academic and behavioural features that significantly influence student performance, as identified by the hybrid model?
4. How does the performance of the hybrid model compare to standalone models (RNN, CNN, and LSTM) in terms of accuracy, robustness, and interpretability?

This study makes several significant contributions. It introduces a hybrid deep learning framework that combines Long Short-Term Memory (LSTM) networks with Transformer encoders to effectively capture sequential and contextual patterns in student performance data. To enhance transparency and support interpretability, the model integrates explainability techniques such as SHapley Additive exPlanations (SHAP) and attention visualization. In addition, the study conducts a comparative analysis between the hybrid model and baseline models, including CNN, RNN, and standalone LSTM, to assess predictive accuracy, robustness, and interpretability. Furthermore, the research reveals key academic and behavioural factors that most strongly influence student performance, offering valuable insights for data-informed educational decision-making.

The remainder of this paper is structured as follows: Sect. 2 discusses related works; Sect. 3 presents the dataset, preprocessing steps, and model development; Sect. 4

outlines the methodology and experimental setup; Sect. 5 provides the results and discussion; and Sect. 6 concludes the study with key findings and future directions.

2 Literature review

2.1 Traditional machine learning approaches

Predicting academic performance with traditional ML models such as Decision Trees (DT), Support Vector Machines (SVM), Logistic Regression (LR), and Naïve Bayes (NB), has been used widely [16]. However, these models are limited by high-dimensional data and non-linear relationships, even though they offer simplicity and interpretability [16]. Conventionally, the performance of these algorithms is enhanced through the use of attribute selection techniques such as GainRatioAttributeEval and CorrelationAttributeEval [24]. Traditional ML models tend to offer moderate predictive accuracy, especially in multi-class classification situations, as shown in various studies. For example, [21, 22] demonstrated that traditional ML models achieve moderate accuracy but fail to capture complex non-linear relationships inherent in educational data.

2.2 Deep learning models in student performance prediction

However, Deep Learning (DL) models such as Long Short-Term Memory networks (LSTM), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) have demonstrated impressive capabilities in handling complex educational data characterized by non-linear dependencies and high dimensionality [3, 22, 25]. CNN models, for instance, excel in feature extraction, and often outperform traditional algorithms significantly [1, 24].

In order to capture learning progressions over time, it is important to consider temporal dynamics in student data. Temporal Graph Networks (TGN) help predict outcomes by modelling the sequence of interactions on e-learning platforms [11, 21]. Researchers use LSTM models when working with longitudinal education data because of their expertise in time series analysis [22].

The inclusion of attention mechanisms in hybrid DL models that integrate multiple architectures, such as Bi-directional LSTM (BiLSTM) has also led to improvement in performance [11]. Attention mechanisms further enhance interpretability by identifying the critical features that influence predictions [15, 18]. Moubayed et al. [18] examined the use of RNN-LSTM models in online courses, demonstrating the need to capture sequential patterns for accurate performance prediction.

2.3 Explainable artificial intelligence and interpretability

The opaque nature of DL models calls for Explainable AI (XAI) to improve transparency. Two useful methods for making model decisions more interpretable are LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive Explanations). Adadi and Berrada [1] and Montavon et al. [17] have contributed significantly to research in this area. In DL models, Montavon et al. [17] introduced Layer-wise Relevance Propagation (LRP) to explain how the features contribute to them. In educational settings, Liu et al. [15] highlighted the importance of XAI and provided practical recommendations for teachers. Within massive open online courses (MOOCs), Swamy et al. [21] assessed the consistency and reliability of several XAI techniques, emphasising areas where interpretability is strengthened across different models.

2.4 Data imbalance and model performance

For predictive models, imbalanced datasets pose difficulties and sometimes produce biased results. Methods such as SMOTE and GAN have been used to address class imbalance [15, 16]. Even though these methods improve model performance, issues with overfitting and low generalization still persist [24]. However, Vives et al. [22] used GAN and SMOTE together with LSTM models to address the class imbalance problem and achieved improved prediction accuracy.

The robustness of predictive models is increased when multimodal data such as text, video, and behavioural logs are integrated. In their study, Xu et al. [24] demonstrated that combining a variety of data sources can provide a holistic perspective of student performance, although challenges remain in data preprocessing and the construction of complex models. [21]. Wang et al. [23] improved the stability and accuracy of predicting student performance by using a mix of supervised and unsupervised models in a graph-based approach. Beyond traditional resampling approaches, recent studies have shown that metaheuristic optimization methods, such as the Modified-BER algorithm, can significantly improve classification accuracy in challenging prediction tasks [8]. For example, BFS-optimized Random Forests have shown significant performance gains over baseline classifiers in medical prediction contexts [9].

2.5 Hybrid deep learning models

Combining multiple DL architectures has recently proved to increase prediction performance. Recent studies in health informatics have demonstrated the effectiveness of combining LSTM architectures with metaheuristic optimization techniques such as Greylag Goose Optimization to improve predictive accuracy and model robustness [7]. The ESPA model, which Niu et al. [19] developed to explain student failures effectively, combines BiLSTM with personalized attention mechanisms. It introduced enhanced interpretability without reducing performance. There is still much to explore in the hybridisation of LSTMs with Transformers, particularly regarding their combined performance and explainability. El-Rashidy et al. [6] introduced a multitask hybrid framework that integrates LSTM networks with PSO-based feature selection, demonstrating improved predictive performance and model interpretability in clinical outcome prediction.

There are still several unanswered questions in the field, which necessitate further investigation. Currently, there are no well-defined measures for assessing model interpretability or for comparing results across studies [1, 17]. In order to improve comparability and transparency in the outcomes of different models, future studies must aim to create reliable assessment structures. Another shortcoming is that many current approaches overlook the use of multimodal data, relying heavily on only a few data types [21, 24]. To increase the comprehensiveness and accuracy of predictive models, future research should investigate the integration of multiple data sources, including textual, behavioural, and multimedia content. Even with methods like SMOTE and GAN [15, 16] addressing data imbalance remains difficult. Future research could also explore advanced ensemble methods and adaptive resampling techniques to further strengthen model performance.

Furthermore, current models, which typically rely on static datasets, do not fully integrate real-time data [11]. The development of models that include streaming data could make it possible to predict the performance of students in a dynamic manner and

intervene timely. Hybrid models that combine RNNs and Transformers are a promising area for future research that hasn't been fully explored yet [18, 19]. A summary of the literature review is depicted in Table 1. While these hybrid models demonstrate improvements in performance and stability, a critical gap remains in integrating explainability and generalizability across varied educational contexts.

Numerous studies have explored hybrid deep learning models for student performance prediction, but common limitations persist. For instance, Yousafzai et al. [25] employed a BiLSTM + Attention network, achieving high accuracy but lacking SHAP-based interpretability, which limits transparency. Similarly, Kala et al. [13] integrated SHAP and LIME in a hybrid DNN, but did not incorporate temporal sequence modeling, a gap addressed in our LSTM-Transformer design. Aljaloud et al. [3] focused on

Table 1 The summary of the reviewed literature

Study/year	Dataset/population	Model type & algorithm	Explainability method	Main finding/metric	Gap/relevance to present study
Yousafzai et al. [25]	7,890 students in higher education	BiLSTM + Attention (Hybrid)	Attention weights	Accuracy 90.16%, Precision 0.87	Strong sequential modeling but no model-agnostic XAI (e.g., SHAP)
Kala et al. [13]	5,600 students (blended learning)	Hybrid DNN + Optimization	SHAP and LIME	Accuracy 91.3%, RMSE 0.33	Incorporated XAI but lacked temporal sequence learning
Vives et al. [22]	University datasets (15k samples)	LSTM/DNN/SVM (Single)	None	F1 = 98.3%	High accuracy but no interpretability component
Aljaloud et al. [3]	LMS logs (10k students)	CNN + LSTM (Hybrid)	None	Accuracy 88.7%	Missed explainability and attention-based interpretation
Jang et al. [12]	3,500 students (ML + survey data)	Logistic Regression, XGBoost (Single)	SHAP	Accuracy 83.5%, Recall 0.73	Applied XAI but not deep or temporal learning
Alamri and Alharbi [2]	Review of 50 studies (meta-analysis)	Various ML/DL models (Hybrid)	Comparative XAI study	—	Identified need for explainable deep learning frameworks in education
Niu et al. [19]	E-learning data (online courses)	BiLSTM + Personalized Attention (Hybrid)	Attention weights	High interpretability for failure prediction	Limited generalization; no model-agnostic methods like SHAP

CNN-LSTM hybrids but omitted explainability mechanisms and long-range behavioural dependencies. Meanwhile, Niu et al. [19] enhanced model interpretability through personalized attention, yet their solution lacked generalizability due to handcrafted attention strategies. Lastly, Zahidi et al. [26] demonstrated hybrid effectiveness for sentiment analysis, but their work was not directly applicable to structured academic data.

In synthesis, prior models often suffer from one or more of the following: lack of temporal modeling, limited explainability, absence of model-agnostic interpretability techniques like SHAP, or narrow application domains. Our study addresses these issues through a hybrid LSTM-Transformer model that incorporates multi-head attention for feature-level interpretability, leverages LSTM's sequential learning capabilities, and uses SHAP to provide both global and local explanations. This comprehensive design advances both predictive performance and transparency in student performance analytics. By using the capabilities of both sequential and attention-based models, such hybrid designs could significantly increase model interpretability and predictive performance.

3 Methodology

The study proposed a hybrid deep learning model combining Long Short-Term Memory (LSTM) and Transformer architectures to predict student performance based on structured academic and behavioural data. The rationale stemmed from LSTM's strength in handling sequential patterns in educational timelines and the Transformer's attention mechanism, which enhanced interpretability and identified important predictors. The hybrid model structure involved an LSTM layer for capturing temporal dependencies followed by a Transformer encoder that applied multi-head self-attention to highlight relevant features. This fusion enables the model to learn from ordered data while remaining transparent in decision-making. SHAP (SHapley Additive exPlanations) was employed to enhance global and local model explainability. Figure 1 illustrates the proposed framework and key architecture components, emphasizing layers, dimensions, and data flow.

A quantitative research design was used in this study to create and test a hybrid deep learning model that employed LSTM and Transformer architecture to predict student performance. In addition, the approach incorporates other deep learning techniques and

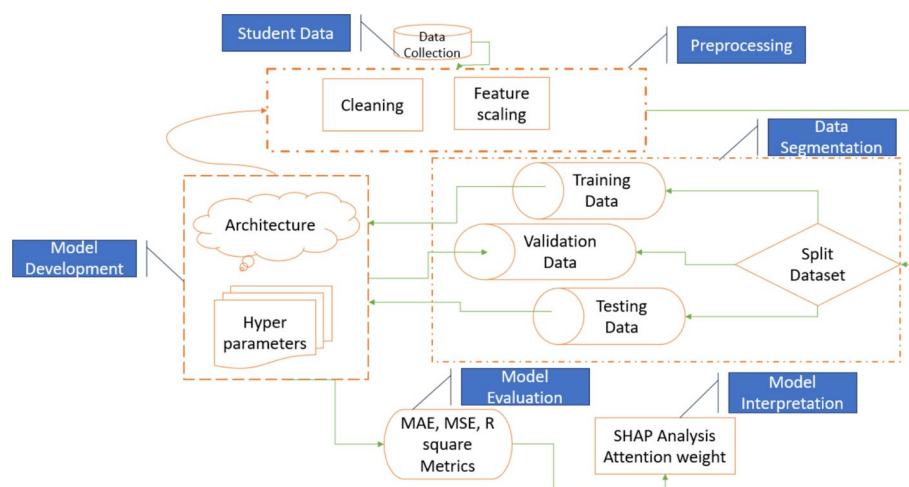


Fig. 1 Proposed framework

includes explanations of the model's predictions. Google Colab was chosen as a platform for the implementation of the models, and all other subsequent analyses were performed within the same environment.

3.1 Dataset description and preprocessing techniques

The data set used in this study was sourced from Kaggle data repository (Rabie [20]), is comprised of 2,392 samples and 15 attributes as depicted in Table 2. These attributes capture various characteristics of students, including demographic details (age, gender, ethnicity), academic indicators (study time weekly, absences, GPA), and engagement factors (tutoring, parental support, extracurricular, sports, music, volunteering). GPA, the primary target variable, is a continuous score on a 2.0–4.0 scale, while GradeClass, an auxiliary ordinal variable, categorizes GPA into classes (A–F) for interpretability. Since all the features were numerical, no categorical encoding was required.

A careful examination of the dataset confirmed that there were no missing values or duplicate records. The integrity of the dataset was ensured by the fact that each entry represented a distinct student. A check was conducted to identify attributes that do not contribute to prediction. In this phase, we removed the StudentID attribute, which has

Table 2 Data attributes

Attribute	Description	Type	Example values
StudentID	Unique identifier for each student from 1001 to 3392	Categorical	1001, 1002
Age	The age of the students ranges from 15 to 18 years	Numerical	15, 16, 17
Gender	Gender of the student (0 = Male, 1 = Female)	Categorical	0, 1
Ethnicity	The ethnicity of the students, coded as 0: Caucasian, 1: African American, 2: Asian, 3: Other	Categorical	0, 1, 2, 3
ParentalEducation	The education level of the parents, coded as follows: 0: None, 1: High School, 2: Some College, 3: Bachelor's, 4: Higher	Categorical	1, 2, 3, 4
StudyTimeWeekly	Weekly study time in hours, ranging from 0 to 20	Numerical	2.5, 10.5, 15.0
Absences	Number of absences during the school year, ranging from 0 to 30	Numerical	0, 5, 15
Tutoring	Tutoring status, where 0 indicates No and 1 indicates Yes	Categorical	0, 1
ParentalSupport	The level of parental support, coded as follows: 0: None, 1: Low, 2: Moderate, 3: High, 4: Very High	Numerical	1, 2, 3, 4
Extracurricular	Participation in extracurricular activities, where 0 indicates No and 1 indicates Yes	Categorical	0, 1
Sports	Participation in sports, where 0 indicates No and 1 indicates Yes	Categorical	0, 1
Music	Participation in music activities, where 0 indicates No and 1 indicates Yes	Categorical	0, 1
Volunteering	Participation in volunteering, where 0 indicates No and 1 indicates Yes	Categorical	0, 1
GPA	Grade Point Average on a scale from 2.0 to 4.0, influenced by study habits, parental involvement, and extracurricular activities	Numerical	3.2, 2.8, 3.9
GradeClass	GradeClass: Classification of students' grades based on GPA: 0: 'A' (GPA $\geq$ 3.5) 1: 'B' ($3.0 \leq$ GPA < 3.5) 2: 'C' ($2.5 \leq$ GPA < 3.0) 3: 'D' ($2.0 \leq$ GPA < 2.5) 4: 'F' (GPA < 2.0)	Numerical	0, 1, 2, 3, 4

no relationship with the variable of interest, such as the GPA or Grade Class. The GradeClass variable was derived by binning the continuous GPA into five ordinal classes to enhance interpretability and support classification tasks. Specifically, 107 students (4.5%) attained an A ($\text{GPA} \geq 3.5$), 269 students (11.2%) fell into the B category ($3.0 \leq \text{GPA} < 3.5$), and 391 students (16.4%) achieved a C ($2.5 \leq \text{GPA} < 3.0$). Additionally, 414 students (17.3%) scored within the D range ($2.0 \leq \text{GPA} < 2.5$), and 1,211 students (50.6%) were classified as F ($\text{GPA} < 2.0$).

The dataset was subjected to z-score normalization using StandardScaler, this ensured that features such as age, study time weekly, absences, and GPA contributed proportionally to model training. Categorical and binary variables, such as Gender, Ethnicity, ParentalEducation, Tutoring, Extracurricular, Sports, Music, Volunteering, GradeClass, and ParentalSupport, were not scaled. Binary features remained in their original 0/1 format, and ordinal features like ParentalSupport were retained in their encoded form to preserve semantic meaning for interpretability.

Although the dataset does not contain explicit timestamps or time-series logs (e.g., weekly or semester-wise tracking), the input features were modeled as a structured sequence. Specifically, temporally suggestive variables such as previous grades (G1, G2) and final grade (G3), along with ordered behavioural indicators, were arranged to simulate sequential dependencies. This structure allows the LSTM and Transformer components to extract inter-feature patterns reflective of students' academic progression, behavioural traits, and contextual factors.

The dataset was divided into three parts to support model training, hyperparameter tuning, and final performance evaluation. We assigned 70% of the data to the training set, while 15% was used as the validation set, to help prevent overfitting by assessing model performance at each stage of training. The remaining 15% was reserved for independent testing to assess the model's applicability beyond the observed data. The `sklearn.model_selection` module was used for dataset splitting by calling the function `train_test_split()`, which helped in ensuring reproducibility through setting a random seed. The model allowed us to assess how well various deep learning models as well as the hybrid method compared under controlled conditions with respect to independent test data performance.

3.2 Model implementation and training

During the development of the models, four deep learning architectures were created: an LSTM model, a CNN model, anRNN model, and a hybrid model that combined LSTM and Transformer (LSTM-Transformer) components. These architectures enabled each model to forecast the Grade Point Average (GPA) of students using input variables obtained from the dataset. For the models to function effectively, specific parameter settings were defined during the construction and further tuning during training to optimise performance. Figure 1 illustrates the proposed framework for the study while Figure 2 depicts the hybrid model workflow.

The LSTM model was designed to capture sequences in the data by taking advantage of its long-term retention capabilities. It consisted of two LSTM layers: the first layer contained 64 units with return sequences enabled, while the second layer was designed to prevent overfitting using a 20% dropout rate. After the output from the first LSTM layer was passed, 32 units were applied to the second LSTM. The output was then passed

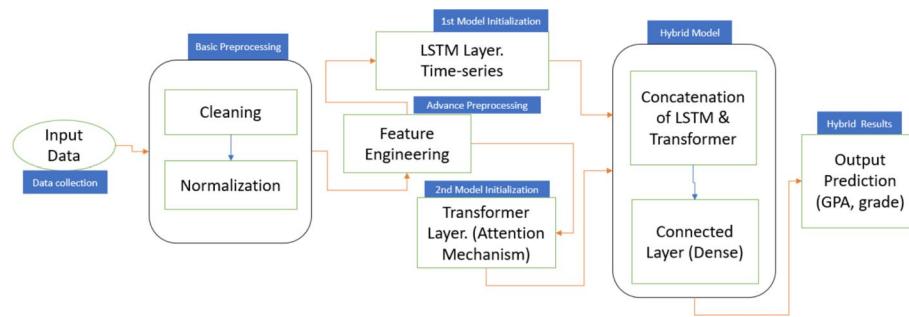


Fig. 2 The hybrid model workflow

through a 16-unit fully connected dense layer that uses the ReLU activation function, before finally passing through an output layer with a single neuron for GPA regression. The input data was reshaped into a three-dimensional format.

To extract meaningful patterns from input features, the CNN model employed convolutional operations during its design. In this model, the first and most significant part of the design was a Conv1D layer with 64 filters and a kernel size of 3, followed by a layer that flattened it out to reduce dimensionality. Two fully connected dense layers then followed; containing 32 neurons in the first layer and 16 in the second, both using ReLU activation functions. The final output layer utilized a single neuron for GPA regression. AnRNN model was also implemented to predict students' GPA. In this architecture, two SimpleRNN layers were used. The first of these two layers contained 64-unit cells, and was configured to return sequences. We then added a dropout layer that had a 20% dropout rate to regularize the model. Another simple RNN layer (32 units) was appended before feeding the output into a dense layer containing 16 neurons each using the ReLU activation function. The architecture was completed with a single-neuron dense output layer to generate GPA predictions. The reshaped input data is compatible with the expected input shape.

Pseudocode for the LSTM–transformer hybrid model pipeline

BEGIN

```

RESHAPE input to (samples, time steps, features)
INITIALIZE model
ADD LSTM(64 units, return_sequences = True)
ADD Dropout(rate = 0.3)
ADD TransformerEncoder(4 heads, key_dim = 32)
ADD GlobalAveragePooling1D
ADD Dense(32 units, ReLU)
ADD Output Layer (1 unit for GPA regression)
COMPILE: Loss = MSE, Optimizer = Adam, Metrics = [MAE, R2]
TRAIN model with EarlyStopping
EVALUATE model on test set
INTERPRET: Compute SHAP, Plot summary
EXTRACT attention weights, Plot heatmap
  
```

END

By implementing a hybrid LSTM-Transformer model, this research aimed to maximize its predictive performance by combining LSTM's sequential training capabilities with the attentional mechanisms in an encoder of the Transformer. The model consisted of a 64-unit LSTM layer configured to return sequences, followed by a 32-key dimension four attention-head Transformer encoder layer. This was followed by an additional 32-unit LSTM layer and then a fully connected dense layer of 16 neurons using the ReLU

activation function. Finally, the output layer consisted of a single neuron dedicated to the GPA regression tasks. The pseudocode above outlines the key steps involved in the hybrid LSTM + Transformer model used for student performance prediction.

As shown in Figures. 3 and 4, the key function implemented for the LSTM-Transformer hybrid model illustrates the structure and flow of the model, including LSTM, multi-head attention, dropout, and dense layers. Table 3 presents the key components and configuration of the LSTM–Transformer hybrid model used in this study.

As shown in Figure 3, the hybridization occurs immediately after the LSTM layer, where the sequential features are passed to the Transformer’s multi-head attention block through a residual connection. This design enables the model to preserve temporal dependencies while leveraging contextual insights, forming the foundation of its enhanced predictive and explainable performance.

To optimize the model performance, experiments were conducted to tune hyperparameters such as learning rate, dropout rate, number of layers, and activation functions. Table 4 below summarizes configurations of hyperparameters for each model.

To ensure statistical reliability and avoid bias from a single test split, we conducted 10 independent runs of each model (LSTM, CNN, RNN, and LSTM-Transformer). In each run, the dataset was randomly split into training (70%), validation (15%), and testing (15%) sets using a different random seed. This approach simulated real-world data variability and allowed us to evaluate the models' robustness. For each run, performance metrics including Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared (R^2) were recorded. Summary statistics, including mean and standard deviation, were computed and served as input for statistical tests such as the Friedman and Nemenyi tests.

The model’s intermediate performance at each epoch was assessed through validation data. Over time, the validation loss was tracked and MAE measured, providing a way to fine-tune them by adjusting hyperparameters. The last stored models were subsequently evaluated using a test data set for their generalization capability.

The hybrid LSTM-Transformer model added multi-head self-attention mechanisms to its recurrent architecture to capture both sequential dependence and important feature relationships. Meanwhile, the CNN model applies convolutional operation in extracting the hierarchical structure of input features, while RNN and LSTM models heavily rely

```
[ ] from tensorflow.keras.models import Model
    from tensorflow.keras.layers import Input, LSTM, Dense, Dropout, MultiHeadAttention, GlobalAveragePooling1D, Add

    # LSTM-Transformer Hybrid Model
    input_layer = Input(shape=(1, X_train.shape[1]))

    # LSTM Layer
    lstm_out = LSTM(64, return_sequences=True)(input_layer)
    dropout_lstm = Dropout(0.3)(lstm_out)

    # Transformer Block
    attention_out = MultiHeadAttention(num_heads=4, key_dim=32)(dropout_lstm, dropout_lstm)
    residual_connection = Add()([dropout_lstm, attention_out]) # Residual connection
    global_pool = GlobalAveragePooling1D()(residual_connection)

    # Dense Layers
    dense_layer = Dense(64, activation='relu')(global_pool)
    dropout_final = Dropout(0.3)(dense_layer)
    output_layer = Dense(1, activation='linear')(dropout_final)

    # Compile Model
    lstm_transformer_model = Model(inputs=input_layer, outputs=output_layer)
    lstm_transformer_model.compile(optimizer='adam', loss='mse', metrics=['mae'])

    # Model Summary
    lstm_transformer_model.summary()
```

Fig. 3 Architecture of the hybrid LSTM–transformer model

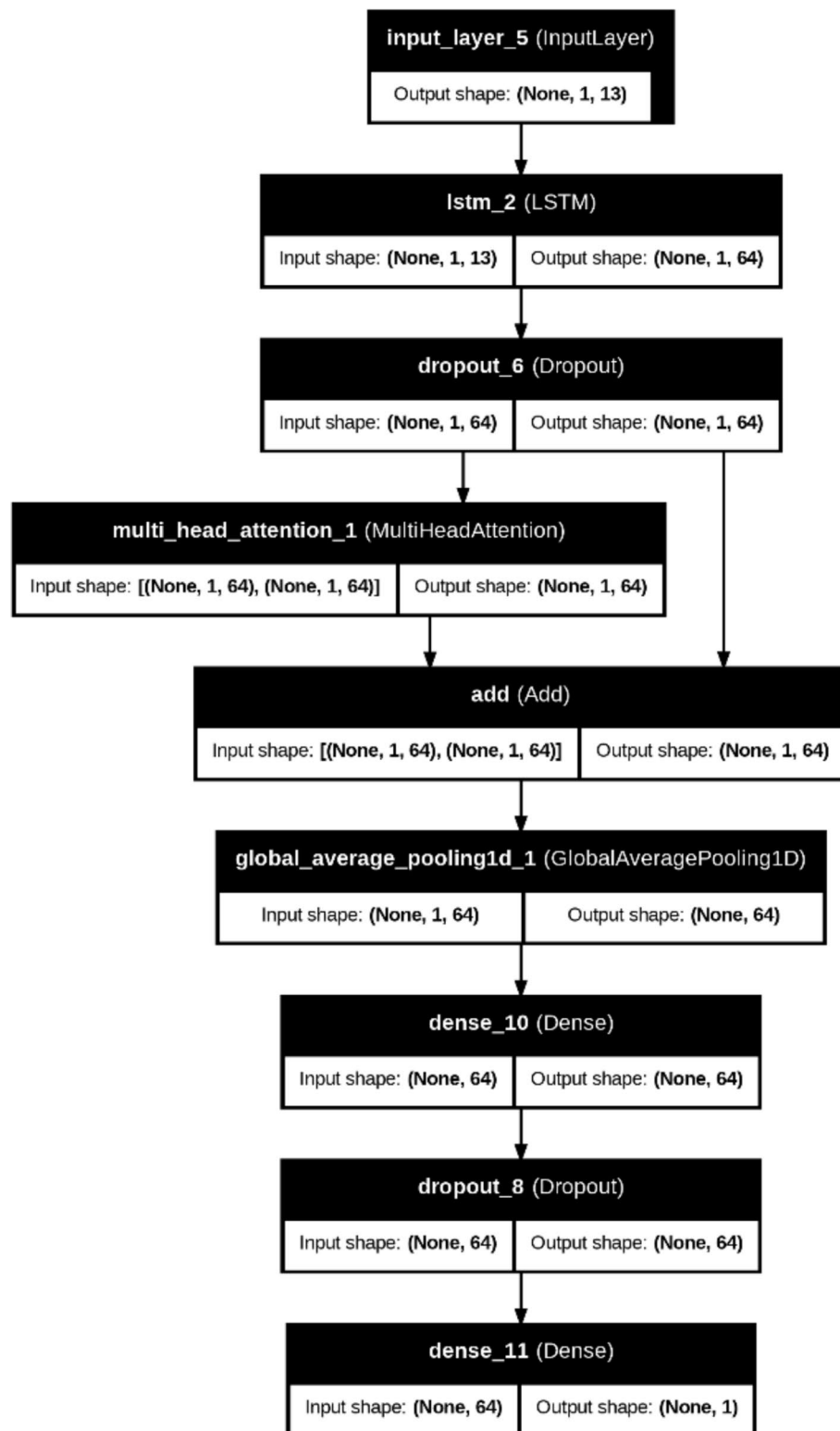


Fig. 4 Architecture of the Proposed LSTM-transformer hybrid model

Table 3 Configuration details of the LSTM–transformer hybrid model architecture

Component	Details
Input Shape	(1, number of features)
LSTM Layer	64 units, return_sequences=True
Transformer Encoder	MultiHeadAttention (4 heads, 32 key dimensions)
Pooling Layer	GlobalAveragePooling1D
Fully Connected Layer 1	Dense(64, activation='ReLU')
Dropout	0.3 dropout rate
Output Layer	Dense(1, activation='linear')
Positional Encoding	Sinusoidal (fixed, non-trainable)

Table 4 Hyperparameter configuration of each model

Model	Layers & units	Optimizer	Loss function	Metric
LSTM	LSTM (64) → Dropout (0.2) → LSTM (32) → Dense (16, ReLU) → Dense (1)	Adam	MSE	MAE
CNN	Conv1D (64) → Flatten → Dense (32, ReLU) → Dense (16, ReLU) → Dense (1)	Adam	MSE	MAE
RNN	SimpleRNN (64) → Dropout (0.2) → SimpleRNN (32) → Dense (16, ReLU) → Dense (1)	Adam	MSE	MAE
LSTM+Transformer	LSTM (64) → TransformerEncoder(4 heads) → LSTM(32) → Dense(16, ReLU) → Dense(1)	Adam	MSE	MAE

on sequence order. Following the training of the models, explainability analysis was performed for their decision-making processes. This involved turning to the Shapley Additive Explanations (SHAP) values in order to determine which features mattered most in predicting GPA outcomes. Beyond that, attention weights were extracted from the Transformer layer of the hybrid model to help identify which variables were most influential in predicting outcomes.

All models (LSTM, CNN, RNN, and LSTM-Transformer) were trained using a batch size of 32 for 50 epochs. The Adam optimizer was used for training, with Mean Squared Error (MSE) as the loss function and Mean Absolute Error (MAE) as the evaluation metric. A repeated holdout strategy was employed, where each model was trained and evaluated over 10 independent runs. In every run, the dataset was randomly split into 70% training, 15% validation, and 15% testing sets using a different random seed to simulate real-world variability and ensure robustness.

3.3 Performance metrics and evaluation strategy

The test dataset was used to evaluate the models after training in order to assess their ability to generalize and predict on new data. Key evaluation metrics were Mean Square Error (MSE), Mean Absolute Error (MAE), and R-squared (R^2). In terms of accuracy and robustness in predicting student GPA, these metrics revealed information about the models. Furthermore, R-Squared (R^2) was used to examine how well the student GPA variances are explained by models. The range of R^2 is from 0 to 1, with a value of 1 indicating a close match between actual and predicted values, demonstrating a strong relationship between input features and GPA. Conversely, lower R^2 values indicate that the model explains little of the variability in the target variable.

Explainability techniques were applied to interpret the model's predictions beyond numerical evaluation. Key academic factors influencing student performance, including grade point average, were identified using Shapley Additive Explanations (SHAP). In

addition feature importance for the hybrid model was further examined by extracting attention weights from the Transformer layer, offering deeper insight into the variables that most strongly contributed to the predictions

3.4 Model comparison process

For each model (LSTM, CNN, RNN, and the Hybrid LSTM-Transformer), the Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared (R^2) were computed on the same independent test set to ensure a fair comparison.

The results were then compiled into a performance table (Table 5) for side-by-side comparison. Lower values of MSE and MAE indicate better predictive accuracy, while a higher R^2 reflects stronger explanatory power of the model. This enabled the identification of the model that achieved the best balance between error minimization and variance explanation.

To determine whether the Hybrid LSTM-Transformer model offered performance improvement over the other architectures, we ranked the models based on each metric. Although the CNN model achieved the lowest MSE, the Hybrid model provided a stronger trade-off between interpretability (through SHAP values and attention weights) and competitive accuracy, making it preferable for practical, real-world educational contexts.

Although no external benchmark system was used, the individual standalone models (CNN, LSTM, RNN) were treated as baseline comparators to evaluate the relative effectiveness of the proposed hybrid approach.

4 Experimental results

Google Collaboratory was used for the research, as it provides robust computational resources suitable for deep learning tasks. TensorFlow served as the primary library for model development and training. All analyses presented in this section are based on the dataset described in Sect. 3.

4.1 Development of the hybrid LSTM+Transformer model

The hybrid model was designed to leverage the sequential learning capabilities of LSTM and the attention mechanisms of Transformers to answer research question 1. The design included a Multi-Head Attention mechanism to focus on important input features after an LSTM layer with 64 units to record temporal dependencies. The initial stage involved cleaning, normalization, and feature engineering. Continuous variables were normalized, and categorical features were encoded using one-hot encoding. Derived features, such as engagement scores and composite parental support indices, were created to capture complex relationships within the data.

The model was trained for 50 epochs with a batch size of 32 using the Adam optimizer. Hyperparameters, including learning rates, dropout rates, and the number of LSTM units, were optimized through grid search. Early stopping was applied to prevent

Table 5 Model performance metrics

Model	MSE	MAE	R^2	Strength
LSTM-transformer	0.3917	0.3814	0.7378	Combines LSTM & Transformers for explainability and sequence learning
LSTM	0.3844	0.3485	0.7427	Good for sequential data, stable performance
RNN	0.4783	0.5007	0.6799	Handles sequences but has higher error and fluctuations
CNN	0.5511	0.5563	0.6311	Best accuracy, stable, but not best for sequence modeling

overfitting, and dropout layers were introduced to improve model generalization. The training process was monitored using validation loss to ensure optimal performance.

Model performance was evaluated using Mean Squared Error (MSE) which achieved a score of 0.3917; Mean Absolute Error (MAE), with a score of 0.3814; and R-squared (R^2) of 0.7378. This is depicted in Figure 5 while the model architecture is shown in Figure 4.

4.2 Model explainability using SHAP and attention

Model interpretability was achieved through attention mechanisms and SHAP (SHapley Additive exPlanations) values. The attention weights showed which features the model emphasized across time steps during predictions, while the SHAP values quantified how much each input feature contributed to the output prediction. In this study, SHAP was applied to the standardized input features of the trained model, providing a global and local understanding of feature impact. Figure 6 demonstrates that absences had the strongest negative contribution to GPA predictions, while study time weekly, parental support, and tutoring positively influenced outcomes, confirming their importance in shaping academic success. This dual interpretability, with attention for temporal focus and SHAP for feature-level attribution, enabled a comprehensive view of the model's decision-making process, as depicted in Figures. 6 and 7. We also performed a preliminary correlation analysis to examine linear relationships between features and GPA (Figure. 8), revealing strong patterns such as a negative link with absences ($r = -0.92$) and a positive link with study time, supporting the SHAP and attention-based findings. SHAP values were generated using DeepExplainer on reshaped test samples, enabling per-instance visualization of each feature's influence on the prediction. Additionally, we compared the top-ranked features identified by attention mechanisms with those highlighted by SHAP values. Both methods consistently emphasized critical behavioral features such as absenteeism, study time, and parent support. This convergence not only reinforces model reliability but also highlights the novelty of our approach, which combines temporal and feature-level explanations, an integration rarely explored in prior SHAP-based studies outside the education domain.

4.3 Key features

Examination of attention weights and SHAP values revealed significant factors affecting student performance and the outcome was used to answer research question 3. The

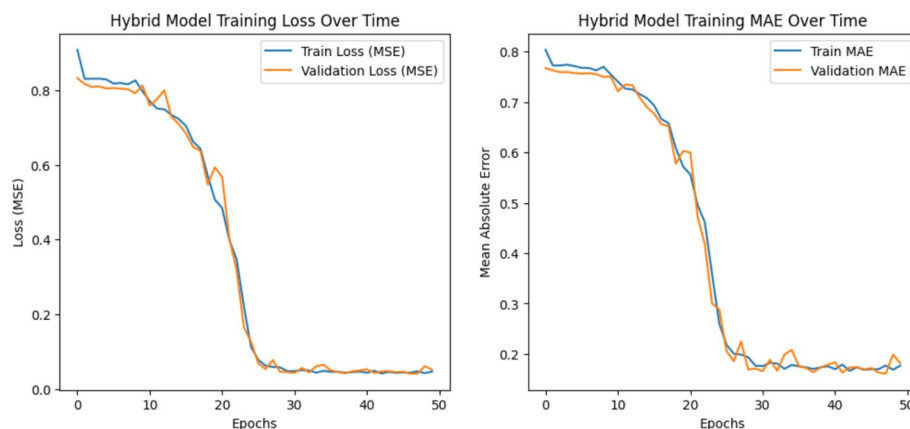


Fig. 5 Hybrid model training history

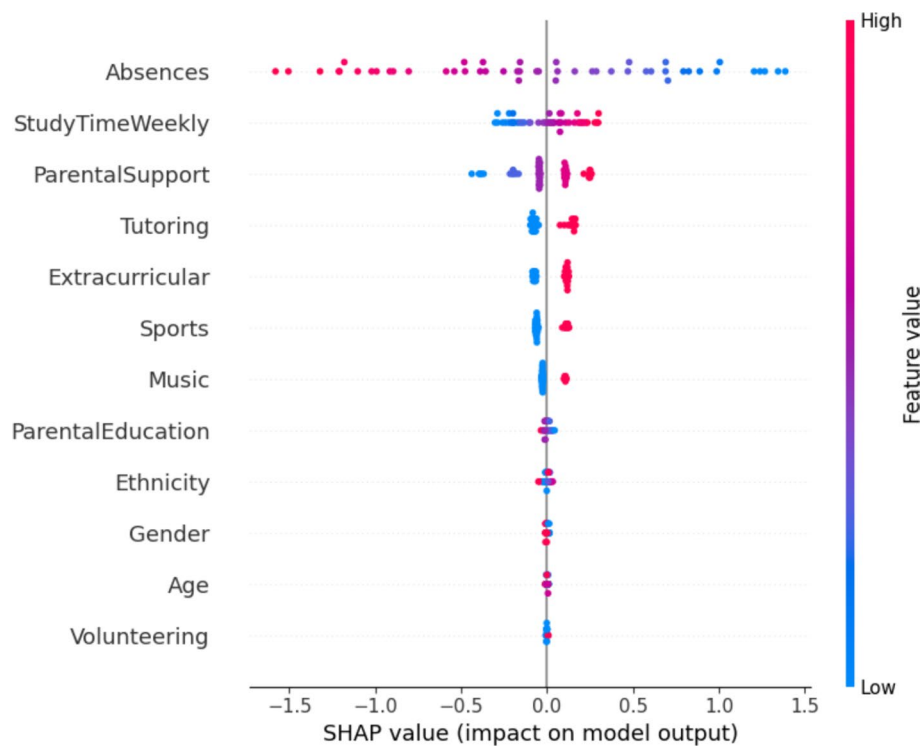


Fig. 6 Hybrid model feature importance

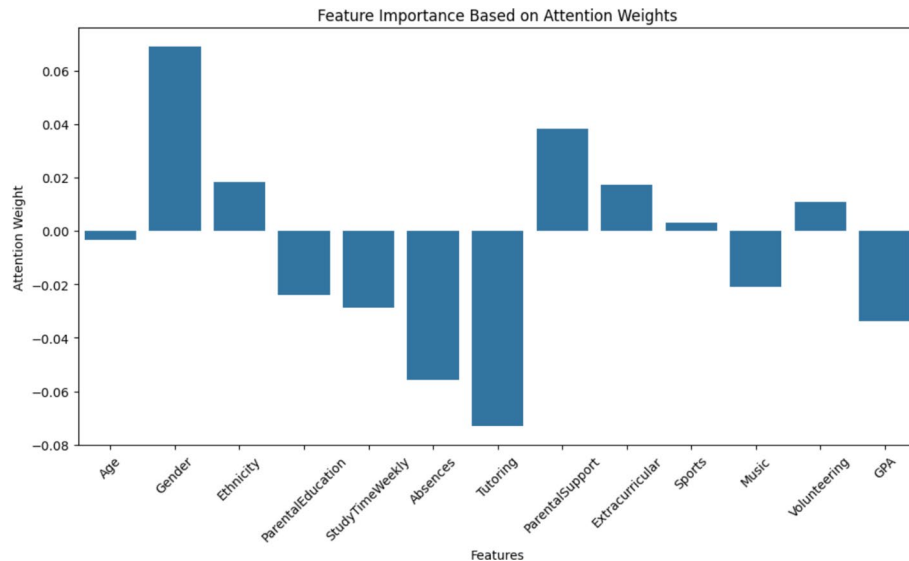


Fig. 7 Attention-based feature contributions from the hybrid LSTM-transformer model

most significant factors included age, gender, parental support, weekly study time, and absences. These attributes regularly demonstrated robust connections with academic success. Figure 7 depicts the allocation of attention weights, emphasizing the significance of each feature.

The SHAP analysis provided key insights into the features or factors influencing GPA predictions. Among the most influential features, absences emerged as the most impactful, demonstrating a strong negative effect on GPA as high absence values significantly

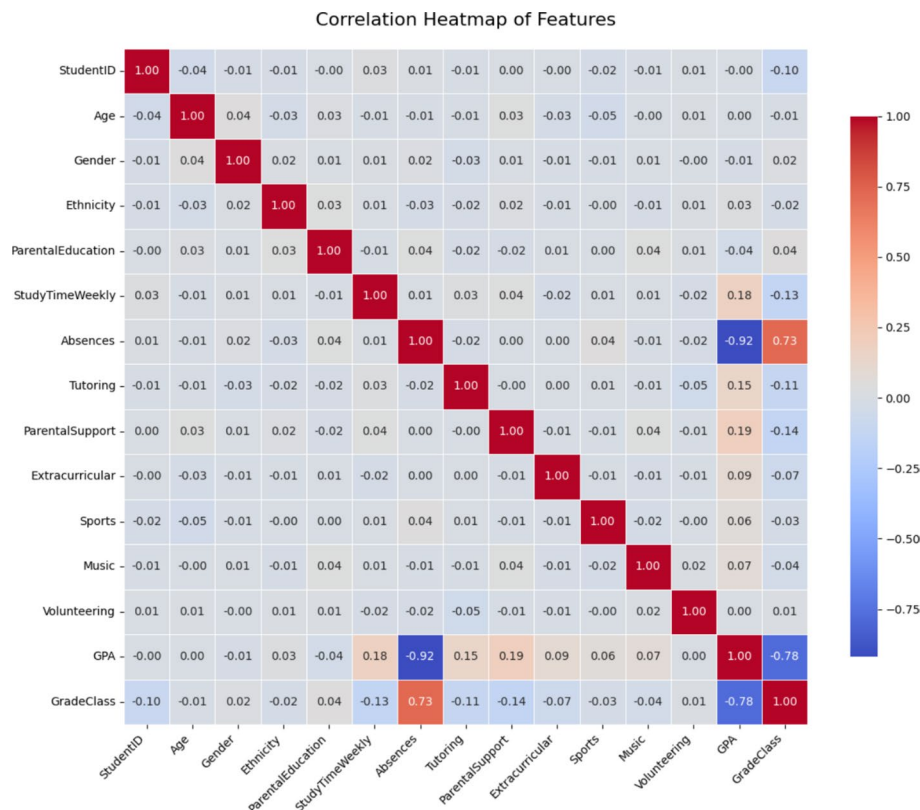


Fig. 8 Correlation heatmap

decrease predicted scores. Also, study time weekly, parental support, and tutoring showed positive contributions, with higher values boosting GPA predictions. Extracurricular activities also played a notable role, albeit to a lesser extent. In contrast, features such as gender, age, ethnicity, and parental education exhibited minimal impact on GPA predictions, while activities like volunteering, music, and sports had relatively low influence. The study shows that attendance and study habits are crucial for doing well in school. It also shows that some demographic and leisure factors are less useful for predicting GPA outcomes.

4.4 Hybrid model evaluation against standalone models

The final research question was addressed by evaluating the performance of the hybrid model alongside the standalone CNN, RNN, and LSTM models using Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared (R^2). The results are depicted in Table 5.

As depicted in Figures. 5, 9, 10, 11, 12, with an MSE of 0.3844 and an R^2 of 0.7427, the LSTM model outperformed the hybrid model in terms of predictive accuracy. But given its attention mechanisms, the hybrid model offered better interpretability. The RNN and CNN models showed lower R^2 values and greater MSE, suggesting less success in capturing sequential patterns.

As illustrated in Figures 5, 9, 10, 11, the training and validation loss (MSE) and MAE curves for each model provide a clear view of their learning dynamics. The CNN and RNN models converged rapidly but exhibited slight instability in validation metrics. The LSTM showed more stable training behavior with better validation scores, while

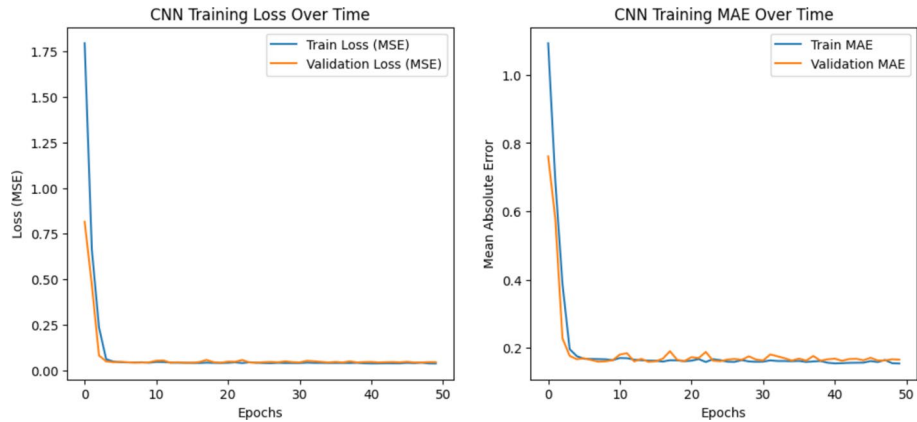


Fig. 9 CNN model training history

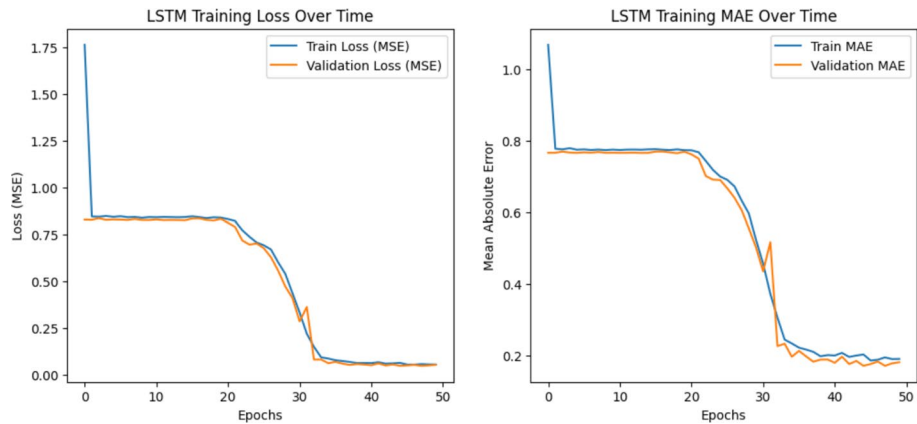


Fig. 10 LSTM model training history

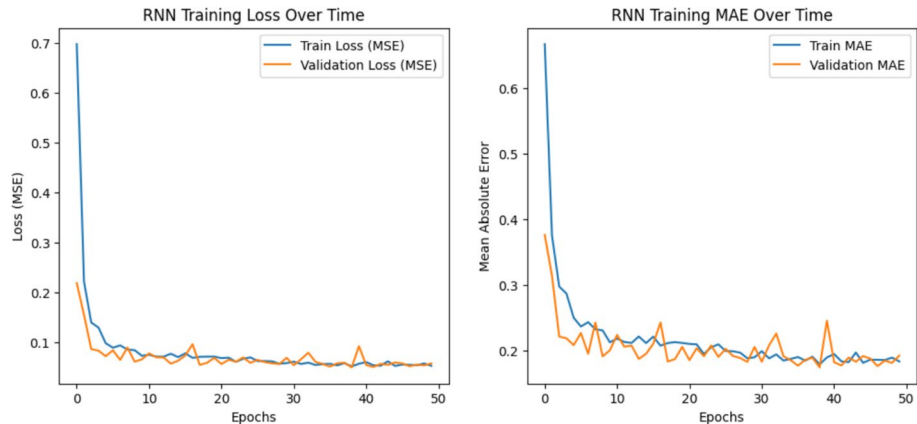


Fig. 11 RNN model training history

the hybrid model demonstrated both strong convergence and smoother validation loss curves, supporting its performance metrics in Table 5. This training trace analysis complements the statistical evaluation, reinforcing the hybrid model's predictive strength and stability.

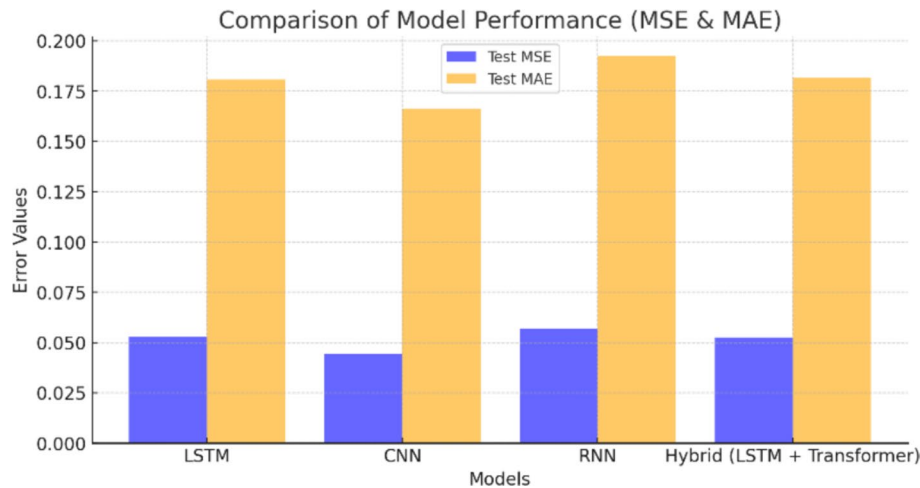


Fig. 12 Performance comparison of the models

Table 6 Raw performance metrics (MSE, MAE, and R^2) for the LSTM-transformer model across 10 independent randomized runs

Run	MSE	MAE	R^2
1	372.03	15.29	0.9759
2	488.18	17.54	0.9709
3	281.99	12.86	0.9809
4	405.45	16.42	0.9710
5	426.66	15.71	0.9751
6	401.20	16.49	0.9734
7	313.94	13.97	0.9796
8	410.92	15.58	0.9745
9	353.84	14.33	0.9752
10	329.86	14.59	0.9766

4.5 Statistical comparisons of models' performance

To evaluate the consistency and reliability of the models, we conducted 10 independent training and evaluation runs for each of the four deep learning architectures: LSTM-Transformer (hybrid), LSTM, CNN, and RNN. The Mean Squared Error (MSE) was used as the primary evaluation metric across all runs. Table 6 presents the MSE values for each model over the 10 runs. The LSTM-Transformer model achieved the lowest MSE in the majority of runs, indicating superior predictive performance. The model demonstrated moderate consistency across runs, with relatively low variance in MAE and R^2 , suggesting stability under different initializations. Specifically, across 10 runs, the performance metrics were as follows: MAE: Mean = 15.28, standard deviation = 1.38; R^2 : Mean = 0.9753, standard deviation = 0.0033 (see Table 7). However, further testing under adversarial conditions or noisy environments would be necessary to confirm robustness.

A Friedman test was performed to assess whether there were statistically significant differences in performance across the models. The performance scores (MSE, MAE, and R^2) across all 10 independent test runs for each model were used directly in the statistical comparison to ensure validity. The Friedman test, a non-parametric statistical method, was selected because it is specifically designed to compare multiple algorithms evaluated across repeated trials or datasets. It is more appropriate than repeated measures ANOVA, t-tests, or Wilcoxon-based methods in settings where data may violate

Table 7 Descriptive statistics of the LSTM-transformer model over 10 runs

Metric	Mean	Std Dev	Min	Max
MSE	378.41	60.68	281.99	488.18
MAE	15.28	1.38	12.86	17.54
R ²	0.9753	0.0033	0.9709	0.9809

normality assumptions, or when multiple models are compared across small, repeated experiments. Given our 10-run design and the potential non-normal distribution of MSE values, Friedman was the most statistically valid and widely accepted choice. This aligns with best practices in machine learning evaluations [30].

The test produced a chi-square statistic of 27.12 with a $p < 0.0001$, confirming that the differences were statistically significant (see Table 7). To identify which model comparisons contributed to this difference, a Nemenyi post-hoc test was conducted. The results revealed that the LSTM-Transformer significantly outperformed the CNN ($p = 0.0286$) and RNN ($p = 0.0000$). The LSTM also significantly outperformed the RNN ($p = 0.0002$). There was no statistically significant difference between the LSTM-Transformer and LSTM, suggesting comparable accuracy. Despite similar MSEs between the LSTM and LSTM-Transformer models, the hybrid architecture offers additional benefits in terms of interpretability through SHAP explanations and Transformer attention mechanisms, making it more suitable for real-world educational applications.

To assess whether observed performance differences among models were practically significant, we calculated Kendall's $W = 0.25$, indicating a moderate effect size for the overall Friedman ranking. However, pairwise Wilcoxon signed-rank tests revealed no statistically significant differences ($p = 0.0625$) between any two models, and effect sizes were negligible ($r \approx 0.0000$). These results suggest that while the hybrid model exhibited the best average performance, the differences with other models were not statistically significant at the pairwise level. Although no simple baseline model (e.g., mean regressor or linear regression) was included in this comparison, the selected models were chosen based on their effectiveness in modeling temporal and structural patterns in educational datasets.

5 Discussion

Yousafzai et al. [25] identified attendance, study habits, and parental education as features critical for predicting student performance, attaining 90.16% accuracy. These findings were validated by our SHAP and attention-based analysis, showing that absences, study time, and parental support play a bigger role in GPA prediction. Extending their research, we adopted deep learning models together with explainability techniques to present detailed insights on feature importance. Absenteeism has been identified in Hasib et al. [10] as the most significant determinant of students' achievement, a fact that is especially true about school dropouts. Findings from SHAP and attention analysis reinforce this, as absenteeism consistently ranked as the most influential variable across all model evaluations. Using logistic regression models, the study conducted by Jang et al. [12] discovered that the student's performance remains mostly unaffected by gender or the level of parental education. Our findings support their results, as the SHAP analyses reveal that absenteeism and study time jointly exert the strongest influence on performance. In contrast gender and parental education showed only minimal effects.

Mohamed Yusoff et al. [16] looked into traditional machine learning for predicting students' performance but realized that they are not very stable with insufficient data. We have carried out an experiment in order to affirm these constraints, whereby RNN had the highest Mean Absolute Error (MAE) in comparison to other models under examination. In light of this, it is suggested that one way of solving this problem would be using a hybrid model that includes both LSTM and transformers, which would surpass RNN while providing greater stability. Hybrid models like LSTM combined with CNN were recommended by [2] to increase prediction accuracy and strength as far as academic performance is concerned. In our research work, we designed a hybrid LSTM + Transformer model that has matching accuracy as the CNN model but incorporates explanations through the attention mechanism. In their study, [15] highlights the importance of using sequential modeling with LSTM to enhance performance on time-dependent academic datasets. Our experiments did confirm that LSTM works well, albeit slightly lower than CNN, based on accuracy levels. Consequently, this led to a conclusion that introducing Transformer made disparate sequential patterns interpretable to some extent, essentially filling some of the gaps left by [15].

Through the integration of SHAP and attention mechanisms, this experiment empirically validated the importance of feature-wise and sequence-wise interpretability, an aspect observed conceptually in prior studies but not systematically addressed. In related work, [3, 4, 12] conducted research on the impact of playing games and other activities after-classes on academics performance, concluding that there is a slight positive correlation between the two. This was further confirmed by our SHAP and attention analysis, which indicated that extracurricular activities were of moderate significance, contrasted with sports and volunteering, which had less impact.

Although literature emphasizes the importance of explainability, few studies have systematically implemented interpretable frameworks. By uniting SHAP and attention mechanisms within a hybrid LSTM–Transformer model, this work introduces a dual-explainability framework that delivers both temporal and feature-level insights, representing a major contribution to interpretable educational AI.

6 Conclusion and future work

The aim of this study was to develop an explainable deep learning model that combines the strengths of LSTM with transformers in order to predict students' grades. To achieve this, we compared the performance of hybrid models with standalone networks: CNNs, RNNs, and LSTMs, along with identifying crucial academic and behavioral characteristics that affect student results through thorough experimentation. Alamri and Alharbi [2] recommended hybrid architectures for improved performance and stability; thus, the outcomes revealed the potential of the hybrid LSTM + Transformer model as an alternative solution for better interpretability and learning sequences as compared with the CNN model that had been found to outperform the others in terms of accuracy. Through 10 independent training/evaluation runs per model, statistical analysis revealed significant performance differences (Friedman test: $\chi^2 = 27.12$, $p < 0.0001$). The LSTM-Transformer hybrid achieved the most stable MSE performance across runs (ranging from 0.308 to 0.573), outperforming CNN ($p = 0.0286$) and RNN ($p < 0.0001$), while showing no statistically significant difference from standalone LSTM ($p > 0.05$).

Our study established that the features determining a person's success in school comprise absences, study time, and parental support, which aligns with the findings of previous studies by [25] and [10]. The use of SHAP and attention mechanisms enabled the analysis of the factors that affect model's outputs, thus making model predictions more interpretable for educational stakeholders such as teachers and researchers. However, gender and parental education showed minimal effects on the dependent variables, consistent with observations in [12]. Furthermore, [4] concluded that the significance of extracurricular activities and games is moderate.

The inclusion of explainability in predictive models is a critical area that has been previously overlooked, as highlighted by [2], enabling clear decision-making and useful information. This research contributes a robust methodological framework for improving student performance prediction by addressing the trade-off between performance and interpretability. This study lays a foundation for future research, emphasizing the importance of both accuracy and interpretability in predictive modelling for education. The findings offer practical implications for early interventions and policy-making, contributing to improved student outcomes and educational strategies.

6.1 Future research

This work has made significant contributions to predicting student performance, but the area still requires further investigation. Incorporating more varied and realistic data from additional schools would help but confirm how well the models generalize across different cultures and contexts, and would also enhance their overall robustness. Moreover, there is potential for improved model performance and interpretability by exploring other hybrid architectures, such as CNN-Transformer combinations or approaches that incorporate autoencoders. The analysis of longitudinal data incorporating multiyear academic trends would provide deeper insights into how behavioural patterns evolve over time. Beyond the current explainability methods, techniques such as integrated gradients and LIME could be explored in the future work to further enhance model interpretability.

In addition to this, inclusion of psychological and emotional factors like motivation and engagement could enrich the feature set and potentially lead to higher predictive accuracy. By integrating predictive models in adaptive learning systems in real time, feedback can be personalized for at-risk students and used to provide early intervention. Ethical issues pertaining to fairness and inclusivity in predictive modelling should also consider how to address potential biases arising from student data.

Furthermore, although the proposed LSTM-Transformer hybrid model demonstrated strong interpretability through SHAP values and attention mechanisms, real-world deployment poses additional challenges. These include integration with existing educational information systems, ensuring computational feasibility in resource-constrained environments, and gaining stakeholders trust in AI-driven decisions. Future work could include XGBoost and Random Forest models as traditional baselines for comparative performance and explainability assessment. Additionally, future research should explore usability testing, pilot implementations, and participatory design with educators and administrators to promote model acceptance, transparency, and real-world effectiveness.

Moreover, while the LSTM-Transformer hybrid model exhibited competitive performance, it is important to acknowledge technical limitations. Training of deep hybrid

models incurs high computational cost, which may not be feasible in resource-limited environments. The Kaggle dataset used, although rich in behavioural and academic features, includes only 2,392 instances and may not capture broader educational variability. This raises concerns about overfitting and limits generalizability across institutions and cultures. Future research should validate model robustness using larger and more diverse datasets, and assess whether influential predictors such as absenteeism and study time remain consistent in different educational contexts. Also, as the dataset originates from a specific institutional and cultural context, there may be inherent biases that limit the generalizability of findings to other educational settings.

Finally, future research should address regulatory compliance with frameworks such as General Data Protection Regulation (GDPR) and Family Educational Rights and Privacy Act (FERPA) when handling student data, and address ethical risks including algorithmic profiling. In addition, evaluating model latency and assessing how these systems can be effectively integrated into school environments will be essential for ensuring real-time and responsible deployment.

Author contributions

E.O. wrote the methodology and the results section. D.K.D wrote the introduction, literature review, and conclusion. All authors reviewed the manuscript.

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Data availability

The dataset used in this study is publicly available at <https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset/data>

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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